

Proposition 1. Let X be a locally compact (i.e., by definition, Hausdorff) space countable at infinity (i.e., countable union of compact subspaces). Let $F \in \mathcal{D}^b(\mathbf{Z}_X)$, and $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . Let $k \in \mathbf{Z}$. If s is a real number, set $Z_s := \bigcap_{t > s} \overline{U_t \setminus U_s}$. We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t \setminus U_s} \cap \text{supp } F$ is compact.
- (iii) For all $s \in \mathbf{R}$ and all $x \in Z_s \setminus U_t$, we have

$$(H_{X \setminus U_s}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For all pairs (s, t) with $s < t$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Proof. Consider the following conditions for $j \in \mathbf{Z}$, s , and t :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t > s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t > s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_s; F) \xrightarrow{\sim} \varprojlim_{r < s} H^j(U_r; F). \end{aligned}$$

Let us prove $(a)_s^j$ for $s \in \mathbf{R}$ and $j < k$. We can assume that $\overline{U_t \setminus U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X \setminus U_s}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_s}(F) \xrightarrow{+1}.$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i: Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\cdot)|_{Z_s \cup U_s} = i_s^{-1}$$

to the d.t. above, we have

$$\text{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s} \rightarrow F|_{Z_s \cup U_s} \rightarrow \text{R}\Gamma_{U_s}(F)|_{Z_s \cup U_s} \xrightarrow{+1}.$$

Again applying to this triangle the functor $\text{R}\Gamma(Z_s \cup U_s; \cdot)$, we have

$$(1) \quad \begin{array}{ccc} & \text{R}\Gamma(Z_s \cup U_s; \text{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) & \\ \nearrow^{+1} & & \searrow \\ \text{R}\Gamma(U_s; F) & \xleftarrow{\quad} & \text{R}\Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}). \end{array}$$

By hypotheses (iii) and (iv), it follows that

$$H_{X \setminus U_s}^j(F) \Big|_{Z_s \cup U_s} = 0 \quad \text{for any } j \leq k,$$

and then

$$H^j(Z_s \cup U_s; \mathrm{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) = 0 \quad \text{for any } j \leq k.$$

Thus for any s , we have

$$(2) \quad H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for any } j < k,$$

$$(3) \quad H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \hookrightarrow H^k(U_s; F).$$

Therefore, if we show Claim 2 below, we have

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and (a)_s^j follows. Similarly, we have

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and (a')_s^k follows.

Claim 2. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}).$$

□

References

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